



Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Software Engineering
2019 Batch

Group B

Attempt any SIX questions

2. What is waterfall model? Describe the activities of waterfall model and also mention its drawbacks.
3. What is CASE? Explain the importance of CASE tools in software development life cycle.
4. Define modularization? Differentiate between cohesion and coupling with suitable example.
5. What are programming languages? Explain various software development tools.
6. Discuss COCOMO model in cost estimation of the software in detail.
7. What do you mean by verification and validation? Differentiate between Black Box testing and White Box testing.
8. Explain activities in software quality assurance with example.

Group C

Attempt any TWO questions.

9. Explain challenges faced by software engineering along with professional and ethical responsibility
10. What is Capability Maturity Model? Explain its levels along with key process area. Differentiate between alpha and beta testing.
11. What do you mean by software maintenance? Explain the different types of maintenance? Explain reengineering process in detail.

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Tribhuvan University
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Semester: IV
Subject: Scripting Language
2019 Batch

Group B

Attempt any SIX questions

2. Explain features of JavaScript?
3. What is NaN in JavaScript? Write JavaScript code to change background color of web page every second.
4. What is arrow function? Write JavaScript code to calculate factorial of given number using arrow function.
5. Write a program to create Chess board in PHP using loop.
6. Write jQuery code to include date picker controls in HTML text box.
7. Write a JavaScript program to retrieve any value from server-side script using Ajax.
8. What is constructor? How you implement constructor in php?

Group C

Attempt any TWO questions.

9. What are advantages of inheritance? Write a PHP program to illustrate the concept of function overloading.
10. Write HTML and JavaScript code to create a following calculator for performing basic arithmetic operations.



11. Design following form in HTML and write corresponding server-side script to upload and submit user's data into database in consideration of following validation rules.

Assume your own database and database connectivity related constrains if necessary.

Validation rules:

- Registration number, email and upload file are mandatory field.
- Email address should not be a proper email format.
- Upload file format must include pdf, doc, docx, ppt, pptx, jpeg file format.
- File size must be less than 5MB.

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Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Database Management System
2019 Batch

Group B

Attempt any SIX questions

2. What is DBMS? Describe the advantages and disadvantages of using DBMS.
3. Describe 3-schema architecture of database.
4. What are roles and responsibilities of DBA.
5. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.
6. What are the states of transaction model? Describe the ACID properties of transaction.
7. Describe the use of ER diagram. Draw an ER diagram for a university to design a database for it.
8. What is DDL and DML queries?

Consider the following relation.

Customer name	Loan number
X	L01
Y	L02

Relation borrower

Loan number	Branch name	Amount
L01	B1	5000
L02	B2	6000

Relation loan

Write a relational algebra expression to find all customer who taken loan from branch "B1".

Group C

Attempt any TWO questions.

9. Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples.
10. Differentiate between stored procedure and trigger. Why concurrency control is needed? Describe timestamp based concurrency protocol.
11. What is data model? What are its types? Describe the structure of database.

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Tribhuvan University
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Semester: IV
Subject: Operating System
2019 Batch

Group B

Attempt any SIX questions

2. What is an operating system? Is "Multitasking" operating system better than "Batch" operating system? Justify your answer.
3. What is role of scheduler in operating system? Differentiate between short-term and long-term Scheduler.
4. What do you mean by Race condition? Consider two concurrently running process P1 with statement S1 and P2 with statement S2. Suppose we require that S2 be executed only after S1 has completed. For this synchronization need, write a semaphore solution.
5. What are CPU scheduling criteria's? Consider below table of four processes under multi-level queue scheduling. Queue number denotes queue of the process. Priority of queue 1 is greater than queue 2. Queue 1 use Round Robin with time quantum $t=2ms$ and queue 2 use FCFS. Draw Gantt chart to illustrate their execution and calculate average waiting and turnaround time also.
6. What is thrashing? Explain three memory management requirements?
7. Given five memory partition of 100KB, 500KB, 200KB, 300KB and 600KB (in order). How would the first fit, best fit and worst fit algorithm place processes of size 212KB, 417KB, 112KB, and 426KB (in order) in fixed-size partition? Which algorithm makes the most efficient use of memory?
8. When programmed I/O is suitable than other I/O handling techniques? Explain different type System Threat.

Group C

Attempt any TWO questions.

9. Consider the following System:

Process	Allocation	Max	Available
	[A,B,C,D]	[A,B,C,D]	[A,B,C,D]
P0	0 0 1 1	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using Banker's algorithm:

- a) What will be the contents of matrix need?
 - b) Draw the safe sequence?
 - c) If the request from P1 Arrives for (0, 4, 2, 0) can the request be granted immediately?
10. How second chance page replacement algorithm differ from FIFO page replacement policy? Explain the concept of belady's anomaly with suitable example.
11. What is clock synchronization? Explain how physical clock is synchronized by Cristian algorithm and logical clock is synchronized by lamport's algorithm with suitable example.

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Semester: IV
Subject: Numerical Methods
2019 Batch

Group B

Attempt any SIX questions

2. If the true value of π is 3.1415926 and its approximate value is given by 3.1428571. Find the absolute and relative errors.
3. What are the advantages of Lagrange's formula over Newton's formula? When Newton's forward interpolation formula is used?
4. Find the polynomial that interpolates $f(0) = 0$, $f(1) = 1$, $f(2) = 0$, $f(3) = 1$, $f(4) = 0$ using Newton divided differences. Use the Newton table to generate the necessary divided differences
5. Using the trapezium rule, evaluate the integral $I = \int_0^1 \frac{dx}{x^2+6x+1}$ with 2 and 4 sub intervals. Compare with the exact solution. Comment on the magnitudes of the errors obtained.
6. For solving a linear system, compare Gauss elimination method and Gauss Jordan method. Why Gauss sieidel method is better than Jacobi's iterative method?
7. Using Euler's method find $y(0.2)$ from $\frac{dy}{dx} = x + y$, $y(0) = 1$ with $h = 0.2$.
8. Solve the Poisson equation $\Delta^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

Group C

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9. When would we not use N-R method? Find the root of the equation $x^3 - 4x + 1 = 0$, lying in $(0, 1)$ Using Bisection method performing 10 iterations.
10. Find the dominant eigen value of $A = \begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix}$ by power method. Apply Gauss sieidel method to solve system of equations.
 $6x_1 - 2x_2 + x_3 = 11$; $-2x_1 + 7x_2 + 2x_3 = 5$; $x_1 + 2x_2 - 5x_3 = -1$, with the initial vector of $(0, 0, 0)$.
11. Prepare the multi-step methods available for solving ordinary differential equation, Evaluate the value of y at $x=0.1$ and 0.2 to 4 decimal places given $\frac{dy}{dx} = x^2y - 1$ $y(0)=1$, using Taylor series method.

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